0.1 Abel's Formular and Abel's Test

0.1.1 Abel's Formular: Summation by Parts

Lemma 0.1.1. Given sequences $\{a_n\}$ and $\{b_n\}$, denote $A_n = \sum_{k=0}^{n} a_k$.
Then, for any $q \in \mathbb{N}$,

$$\sum_{n=0}^{q} a_n b_n = \sum_{n=0}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q = \sum_{n=0}^{q-1} \left(\sum_{k=0}^{n} a_k \right) \cdot (b_n - b_{n+1}) + b_q \cdot \sum_{n=0}^{q} a_n.$$

In particular, if $0 \le p \le q$,

$$\sum_{n=p}^{q} a_n b_n = \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

Proof. Since $a_n = A_n - A_{n-1}$ (with $A_{-1} \overset{\text{def}}{=} 0$),

$$\begin{aligned}
\sum_{n=0}^{q} a_n b_n &= \sum_{n=0}^{q} (A_n - A_{n-1}) b_n = \sum_{n=0}^{q} A_n b_n - \sum_{n=0}^{q} A_{n-1} b_n \\
&= \sum_{n=0}^{q} A_n(b_n - b_{n+1}) + \sum_{n=0}^{q} A_n b_{n+1} - \sum_{n=0}^{q} A_{n-1} b_n \\
&= \sum_{n=0}^{q} A_n(b_n - b_{n+1}) + A_q b_{q+1} \\
&= \sum_{n=0}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q.
\end{aligned}$$

Particular,

$$\begin{aligned}
\sum_{n=p}^{q} a_n b_n &= \sum_{n=0}^{q} a_n b_n - \sum_{n=0}^{p-1} a_n b_n \\
&= \sum_{n=0}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - \left(\sum_{n=0}^{p-2} A_n(b_n - b_{n+1}) + A_{p-1} b_{p-1} \right) \\
&= \sum_{n=p-1}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_{p-1} \\
&= \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_{p-1} b_{p-1} - A_{p-1} b_p + A_q b_q - A_{p-1} b_{p-1} \\
&= \sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.
\end{aligned}$$

□

Note: Denote $A_n = \sum_{k=0}^{n} a_k$ and $\Delta b_n = b_{n+1} - b_n$, then

$$\sum_{n=0}^{q} a_n b_n = A_q b_q - \sum_{n=0}^{q-1} A_n \Delta b_n$$

It is similar to:

$$\int f \cdot g = Fg - \int F g'$$

0.1.2 Abel's Test

> **Theorem 0.1.2.** If $\sum a_n$ converges and $\{b_n\}$ is bounded and monotone sequence, then $\sum a_n b_n$ converges.

Proof. Since $\{b_n\}$ is bounded and monotone, it is convergent. Put $b = \lim_{n\to\infty} b_n$ and $A_n = \sum_{k=0}^{n} a_k$, $A_{-1} = 0$.
WLOG, suppose $\{b_n\}$ is monotone decreasing. Then $\{b_n - b\}$ converges to zero and is decreasing.
Let $M > 0$ be a bound of $|A_n|$. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ such that $n \geq N \implies |b_n - b| < \dfrac{\varepsilon}{2M}$.
By Abel's formula, for all $q \geq p \geq N$,

$$\begin{aligned}
\left|\sum_{n=p}^{q} a_n(b_n - b)\right| &= \left|\sum_{n=p}^{q-1} A_n(b_n - b_{n+1}) + A_q(b_q - b) - A_{p-1}(b_p - b)\right| \\
&\leq M \cdot \left[\sum_{n=p}^{q-1} \underbrace{|b_n - b_{n+1}|}_{\geq 0} + \underbrace{|b_q - b|}_{\geq 0} + \underbrace{|b_p - b|}_{\geq 0}\right] \\
&= M \cdot \left[\sum_{n=p}^{q-1} (b_n - b_{n+1}) + (b_q - b) + (b_p - b)\right] \\
&= M \cdot [(b_p - b_q) + (b_q - b) + (b_p - b)] \\
&= 2M(b_p - b) < 2M \cdot \frac{\varepsilon}{2M} = \varepsilon
\end{aligned}$$

Therefore, by the Cauchy Criterion for series, $\sum a_n(b_n - b)$ converges.
Moreover, since $\sum a_n$ converges, so is $b \cdot \sum a_n$. Finally,

$$\sum_{n=0}^{\infty} a_n(b_n - b) + \sum_{n=0}^{\infty} a_n \cdot b = \sum_{n=0}^{\infty} a_n b_n$$

converges as a sum of two convergent series. □

Note: If $b = 0$, it is sufficient for the convergence of $\sum a_n b_n$ that $\sum a_n$ be bounded.
In this case, $\sum a_n$ does not necessarily have to converge (it is called Dirichlet's Test).

Example. A Series

$$\sum_{n=1}^{\infty} \frac{\sin n}{n}$$

converges. Enough to show that: $\displaystyle\sum_{n=1}^{N} \sin n$ is bounded. Then we can apply Dirichlet's Test. Observe that

$$\begin{aligned}
2\sin(1) \cdot \sum_{n=1}^{N} \sin n = \sum_{n=1}^{N} 2\sin(1)\sin n &= \sum_{n=1}^{N} [\cos(n-1) - \cos(n+1)] \\
&= \cos 0 - \cos 2 + \cos 1 - \cos 3 + \cos 2 - \cos 4 + \cdots + \cos(N-2) - \cos(N) + \cos(N-1) + \cos(N+1) \\
&= \cos 0 + \cos 1 - \cos(N) - \cos(N+1)
\end{aligned}$$

Therefore, for any $N \in \mathbb{N}$,

$$\left|\sum_{n=1}^{N} \sin n\right| = \left|\frac{\cos 0 + \cos 1 - \cos(N) - \cos(N+1)}{2\sin(1)}\right| \leq \frac{2}{\sin(1)}$$

Now, Applying the Diriclet's Test, we obatain the result. But, it is not converges absolutley.

Dirichlet-Dedekind.

Let $f_n, g_n : X \to \mathbb{C}$ $(n \in \mathbb{N})$ be sequence of functions.

Suppose that

a). $F_n(x) \overset{\text{def}}{=} \sum_{k=1}^{n} f_k(x)$ is uniformly bounded.

b). $\sum |g_n - g_{n-1}|$ converges uniformly.

c). $g_n \to 0$ uniformly

Then, $\sum f_n g_n$ converges uniformly.

Proof. By the Abel's formula, given $x \in X$, and $0 < p < q$,

$$\begin{aligned}
\sum_{n=p+1}^{q} f_n(x) g_n(x) &= \sum_{n=1}^{q} f_n(x) g_n(x) - \sum_{n=1}^{p} f_n(x) g_n(x) \\
&= F_q(x) \cdot g_q(x) - \sum_{n=1}^{q-1} F_n(x) \cdot (g_{n+1}(x) - g_n(x)) \\
&\quad - \left[F_p(x) g_p(x) - \sum_{n=1}^{p-1} F_n(x) \cdot (g_{n+1}(x) - g_n(x)) \right] \\
&= F_q(x) \cdot g_q(x) - F_p(x) \cdot g_p(x) - \sum_{n=p}^{q-1} F_n(x) \cdot [g_{n+1}(x) - g_n(x)]
\end{aligned}$$

Now, for bound $M > 0$ of $\{F_n\}$,

$$\begin{aligned}
\left| \sum_{n=p+1}^{q} f_n(x) g_n(x) \right| &= \left| F_q(x) g_q(x) - F_p(x) g_p(x) - \sum_{n=p}^{q-1} F_n(x) [g_{n+1}(x) - g_n(x)] \right| \\
&\leq M \cdot \Big[|g_q(x)| + |g_p(x)| + \underbrace{\sum_{n=p}^{q-1} |g_{n+1}(x) - g_n(x)|}_{\text{Cauchy, by b)}} \Big] \to 0.
\end{aligned}$$

$\blacksquare$